BUSINESS INTELLIGENCE CAPSTONE · COMMERCE

Retail Orders: where revenue is made, and where it leaks

Executive Decision Report for the C-suite · Decision owner: Sales Manager

DATA
retail-orders-raw · 12 records · 3–18 Jan 2026

SCOPE
KPI_Dictionary metrics, revenue drivers, risks, 3 recommendations, ROI

COMPANION
Retail_Executive_Analysis.ipynb

EXECUTIVE SUMMARY

We book ₹13.1k but collect only 46% of it — and fixing collection is the fastest win

₹13,059
Booked revenue (7 clean orders)

₹6,013
Collected (Paid)

57%
Payment success rate

58%
Raw records usable as-is

- **Revenue driver:** Course Access is 65% of booked revenue, but only 18% of it is collected. Learning Kit is smaller and 100% collected.
- **Risk:** every Pending and Refunded rupee (₹7,045) sits in Course Access; 5 of 12 raw rows were duplicates or failed validation, and all 3 Mentor Session orders were among them.
- **Decision:** (1) close the payment gap, (2) gate data at capture, (3) bundle Learning Kit with Course Access. Base case ≈ **₹95,956** a year for ≈ ₹35,000 invested.

Directional: the clean sample is 7 orders over 16 days. Projections extrapolate that run-rate and use stated planning assumptions.

CONTEXT & SCOPE

The question: where is revenue made, where is it leaking, what do we do in 90 days?

THE DATASET

Dataset	Retail Orders (Commerce)
Grain	One row per order record
Size	12 records × 9 columns
Window	3 – 18 Jan 2026
Owner	Sales Manager (weekly refresh)

CATEGORIES & SEGMENTS

Learning Kit · Course Access · Mentor Session
Student · Fresher · Professional
Cities: Chennai, Bengaluru, Hyderabad, Delhi (+ Pune, Mumbai, Kochi in rejected rows)

KPIS GOVERNED BY THE KPI_DICTIONARY

Total Revenue	Qty × Unit Price × (1 − Disc%/100)
Total Orders	Distinct Order ID
Average Order Value	Revenue ÷ Orders
Quantity Sold	SUM(Quantity)
Average Discount	AVERAGE(Discount %)
Paid / Pending Orders	Count by status
Payment Success Rate	Paid ÷ Orders × 100

METHODOLOGY

Every number comes from a rules-based pipeline against the data dictionary

12
Raw records
Cover sheet and CSV agree: 12 × 9

−1
De-duplicate
RT-1004 appeared twice; first kept

~
Normalise & repair
Case fixed; “two” → 2; missing discount → 0% (flagged)

−4
Validate vs dictionary
Date, city, quantity, discount rules; failures quarantined

7
Clean orders
KPIs computed only on these

PRINCIPLES

- Quarantine, don’t guess: rows that break a rule are excluded and listed, not silently fixed.
- Repair only when unambiguous and log it.
- Sensitivity-test the one judgement call (missing discount).

SENSITIVITY RESULT

Setting RT-1003’s missing discount to the worst observed 15% instead of 0% moves booked revenue by only **₹225** (−1.7%). Conclusions hold.

DATA QUALITY

Only 7 of 12 raw rows are decision-grade — and the loss hides the entire Mentor Session line

ORDER	ISSUE FOUND	ACTION
RT-1002	Inconsistent text case (segment/status)	Normalised
RT-1003	Inconsistent text case (segment/status)	Normalised
RT-1003	Discount missing	Set to 0% (justified, flagged)
RT-1004	Duplicate order_id (exact copy)	Removed
RT-1005	city missing	Quarantined (excluded from KPIs)
RT-1006	order_date missing/invalid; quantity not a positive whole number	Quarantined (excluded from KPIs)
RT-1007	discount_pct outside 0-100	Quarantined (excluded from KPIs)
RT-1008	Quantity typed as text ('two')	Repaired to 2
RT-1011	order_date missing/invalid	Quarantined (excluded from KPIs)

5 of 12
rows removed or quarantined

₹4,795
order value at list price we cannot see in KPIs

3 of 3 Mentor Session orders were rejected, so that category has no visible performance at all.

KPI SCORECARD

Headline KPIs look healthy until payment status is applied

Total Revenue
₹13,059
after discount

Total Orders
7
distinct

Avg Order Value
₹1,866
revenue ÷ orders

Quantity Sold
12
units

Avg Discount
4.3%
3 of 7 orders discounted

Paid Orders
4
₹6,013 collected

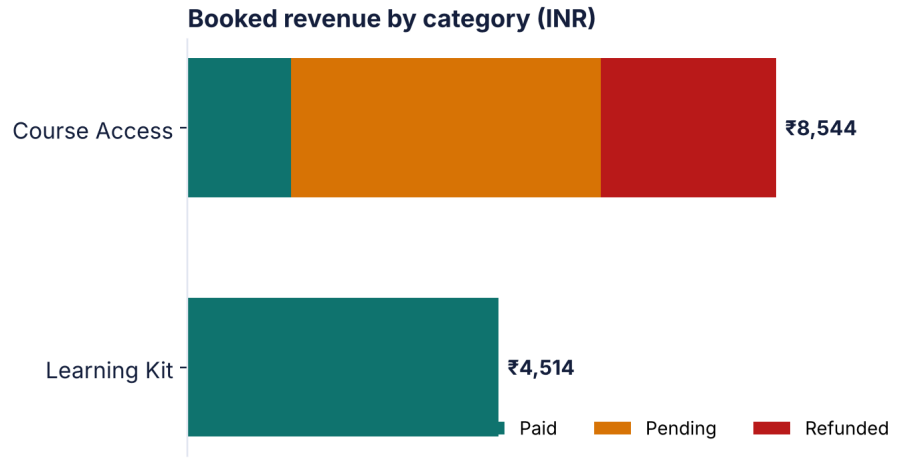
Pending Orders
2
₹4,497 outstanding

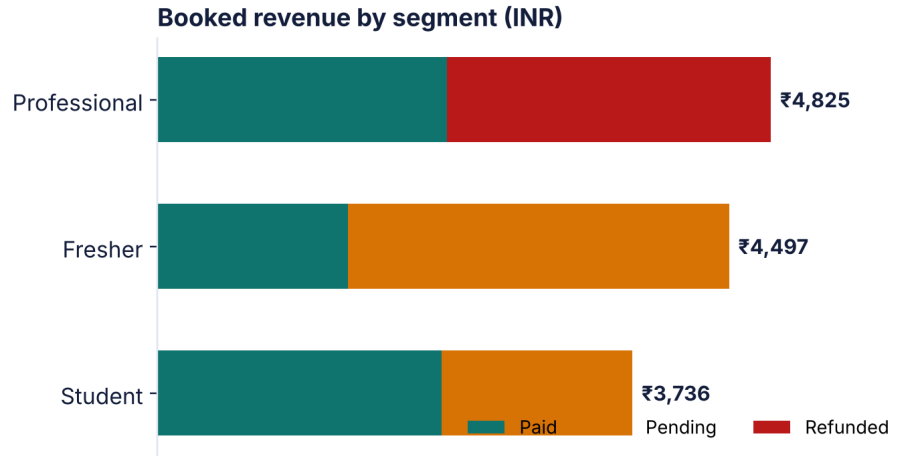
Payment Success
57.1%
paid ÷ orders

Also outside the dictionary but decisive: ₹2,548 refunded (1 order) and a booked-to-collected realisation of 46%.

KEY REVENUE DRIVERS

Course Access drives two-thirds of revenue; Learning Kit is the reliable collector



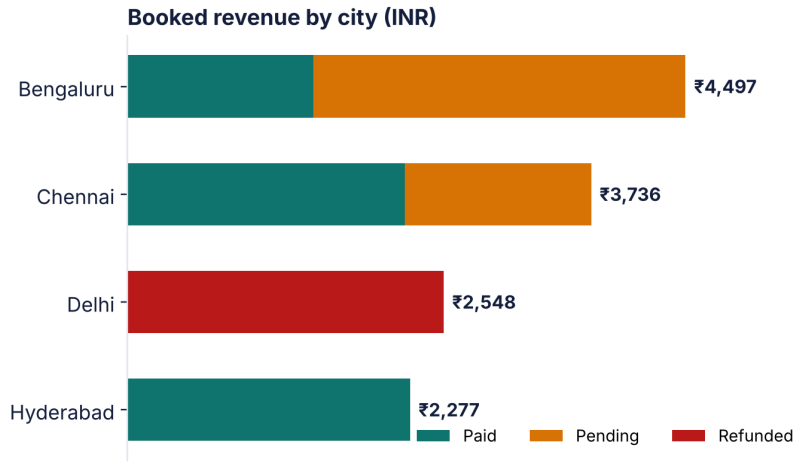


WHAT THIS TELLS US

- **Course Access:** ₹8,544 booked, only ₹1,499 paid.
- **Learning Kit:** ₹4,514, every order paid — a low-friction entry product.
- **Segments are balanced** (Professional 37%, Fresher 34%, Student 29%) but quality differs: Students collect 60%, Fresher 33%.

GEOGRAPHIC VIEW

The biggest city is also the biggest collection gap



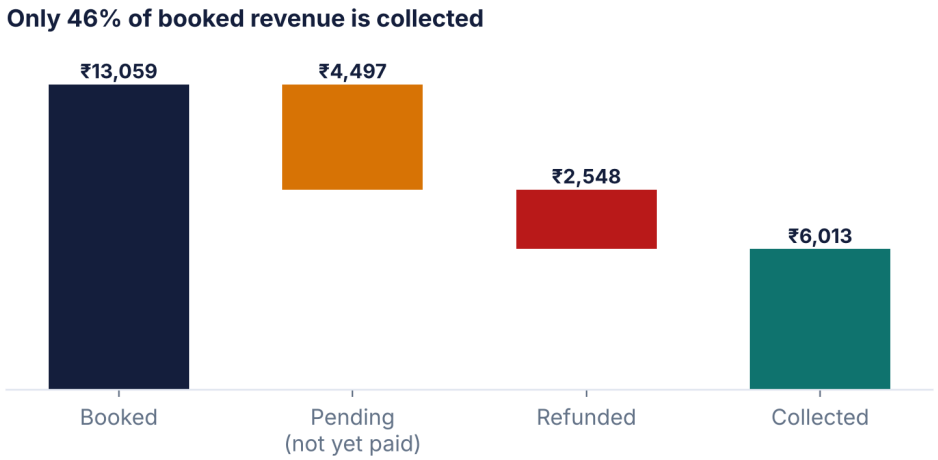
WHAT THIS TELLS US

- **Bengaluru** leads with 34% of revenue but only 33% is collected — both its orders include Pending.
- **Hyderabad** and **Chennai** collect 100% and 60%.
- **Delhi** is a single 15%-discounted Course Access order that was refunded.
- Four cities cannot support a regional strategy yet; treat as a watch-list.

So what: prioritise payment follow-up in Bengaluru first (₹2,998 Pending) before investing in new demand there.

PAYMENT HEALTH

Booked revenue shrinks from ₹13.1k to ₹6.0k before it reaches the bank



34%
of booked revenue still Pending

20%
refunded

Both leaks are Course Access. The Pending value (₹4,497) is recoverable; the refund is a signal to review discount and fit.

So what: every ₹1 of Pending recovered adds straight to collected revenue with no extra acquisition cost. Recovering half the Pending pool lifts collection from 46% to 63%.

RISK AREAS

Five risks, two of them urgent

HIGH	**Collection risk**	₹4,497 Pending (34%) plus ₹2,548 refunded — 100% in Course Access.
HIGH	**Data-integrity risk**	42% of raw rows failed or duplicated. Invalid discount (105%), negative quantity, impossible dates and missing city entered the system.
MEDIUM	**Concentration risk**	Course Access = 65% of revenue; Bengaluru = 34%; a handful of orders moves every KPI.
MEDIUM	**Blind spot**	Mentor Session: 3 raw orders, 0 usable. Performance of a whole product line is unknown.
WATCH	**Discount governance**	3 of 7 orders discounted (5–15%); the one refunded order carried the highest discount (15%). One entry showed 105%.

STRATEGIC RECOMMENDATIONS

Three moves, each tied directly to a finding

1

Close the payment gap

Automated reminders at 24h/72h, payment-link retries and a daily Pending queue owned by Sales Ops. Prioritise Course Access and Fresher segment.

Targets the ₹4,497 Pending pool.

2

Gate data at capture

Hard validation in the order form: unique order_id, date range, city required, whole-number quantity, discount 0–100 with approval above a cap. Weekly DQ report to the owner.

Restores visibility of Mentor Session and ≈ ₹4.8k of unvalidated value.

3

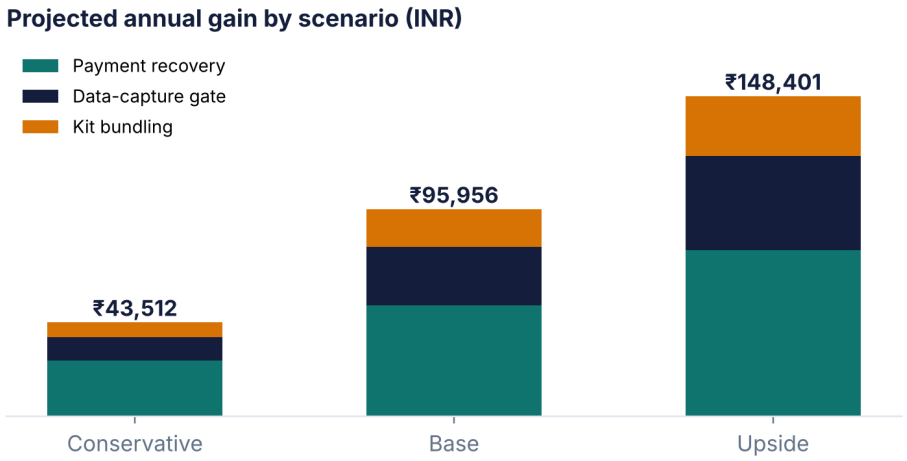
Bundle Learning Kit with Course Access

Offer the 100%-collected kit as an add-on at checkout, with a discount cap (≤10%) and a refund-reason field on discounted Course Access orders.

Lifts AOV and shifts mix toward reliable revenue; test before scaling.

ROI PROJECTIONS

Base case: ≈ ₹95,956 annual gain on ₹35,000 invested, ~4-month payback



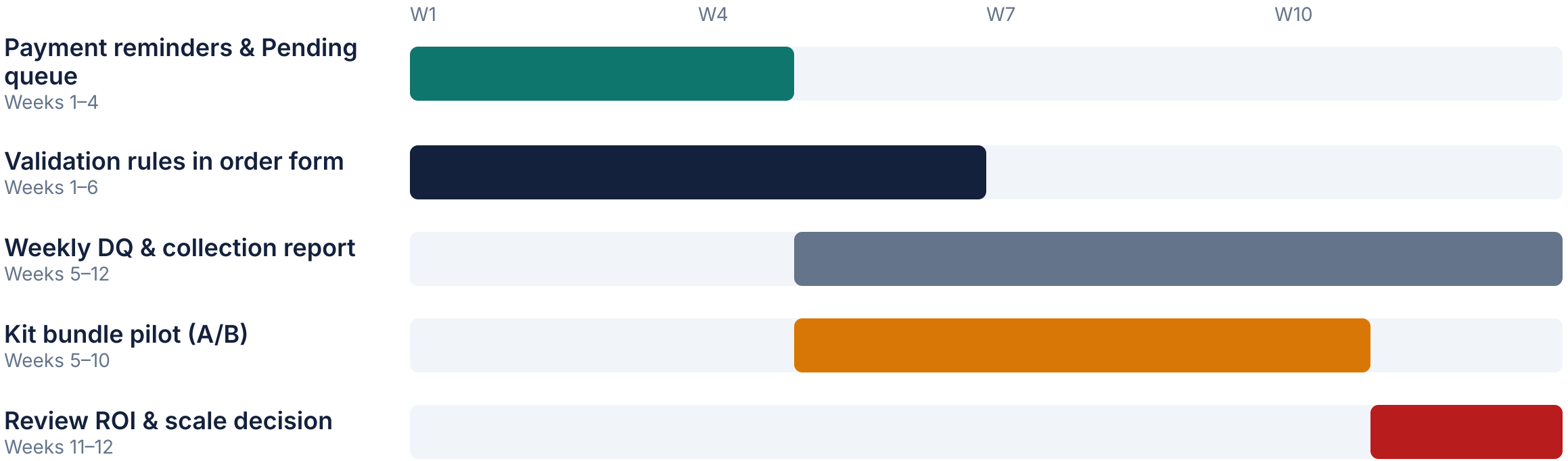
SCENARIO	ANNUAL GAIN	COST	ROI	PAYBACK
Conservative	₹43,512	₹35,000	24%	9.7 mo
Base	₹95,956	₹35,000	174%	4.4 mo
Upside	₹148,401	₹35,000	324%	2.8 mo

ASSUMPTIONS (EDITABLE IN NOTEBOOK)

- Annual run-rate = sample booked revenue ÷ 16 days × 365 = ₹297,900.
- Payment recovery: 25% / 50% / 75% of Pending recovered.
- Data gate: 10% / 25% / 40% of quarantined value becomes valid revenue.
- Bundling: 10% / 25% / 40% of Course Access orders add a kit.
- Costs (planning placeholders): ₹12k reminders/retries, ₹15k validation, ₹8k bundle set-up. Replace with real quotes.

IMPLEMENTATION TIMELINE

A 90-day plan with a go/no-go at week 12



Success measures: payment success rate ≥ 80%, realisation ≥ 70%, raw-to-clean pass rate ≥ 95%, Mentor Session visible in weekly KPIs.

ASSUMPTIONS, LIMITATIONS & NEXT STEPS

What to keep in mind before acting

LIMITATIONS

- 7 clean orders over 16 days: percentages swing by 14 points per order. Use as direction, not forecast.
- “Total Revenue” follows the KPI_Dictionary (all statuses). Collected revenue is shown separately.
- RT-1003 discount set to 0% by assumption; effect ≤ ₹225.
- Annualisation ignores seasonality; costs are placeholders.

NEXT STEPS

- Re-run the notebook on a full month of orders (swap the workbook, run all cells).
- Obtain true costs and Pending-aging data to firm up ROI.
- Ask the owner to confirm the 4 quarantined orders so they can be restored.

DELIVERABLES

This PDF · Retail_Executive_Analysis.ipynb · Retail_KPI_Workbook.xlsx · charts/